A Comparative Analysis of LPWAN Technologies for Smartphone “Disaster Mode”

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Abstract

Natural disasters often disrupt terrestrial communication and power infrastructure, leaving affected populations unable to send or receive critical distress signals. To address this gap, we propose a smartphone-integrated "Disaster Mode" that activates when conventional networks fail and automatically switches to Low-Power Wide-Area Networks (LPWANs) such as LoRaWAN, Sigfox, and NB-IoT. These technologies are designed for low-data, long-range, and energy-efficient communication, making them suitable for emergency use. We analyze how each LPWAN meets the stringent requirements of disaster scenarios and examine the role of satellite connectivity as a final fallback to ensure message delivery in total network blackouts. The paper also discusses current implementations like BSNL’s satellite-backed NB-IoT in India and identifies key technical and policy measures necessary to make Disaster Mode a reliable, scalable national standard.

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I. INTRODUCTION:

Mobile phones play a vital role in emergency communication, often serving as the only means for individuals to request help during natural disasters and crises. However, conventional cellular and Wi-Fi networks frequently become unreliable or unavailable in such events due to damaged infrastructure, network congestion, or power outages. Ensuring continuous, resilient communication during disasters is thus a critical challenge aligned with global and national sustainability goals focused on disaster preparedness, digital inclusion, and energy efficient technology. For example, United Nations Sustainable Development Goals (SDGs) such as Goal 9 (Industry, Innovation, and Infrastructure) and Goal 11 (Sustainable Cities and Communities) emphasize building resilient communication networks and reducing the impact of disasters on communities. In response to these challenges and goals, we propose a novel Disaster Mode for smartphones that automatically switches to a Low-Power Wide-Area Network (LPWAN) link when traditional cellular connectivity fails. Embedding this mode within smartphones removes the need for dedicated external emergency devices, which can be costly, prone to loss, or require separate maintenance.

The Disaster Mode enables essential emergency messaging, such as SOS alerts and GPS location pings, over long distances with minimal battery consumption, making it practical for real-world disaster scenarios where power is limited and rapid response is critical. The primary use cases for Disaster Mode include natural disasters (earthquakes, floods, storms) and other emergency situations where cellular networks collapse or are overloaded. The system must reliably deliver brief status messages with the following requirements: wide-area coverage (kilometer-scale range), ultra-low power operation to preserve limited battery life, support for very low data rates due to short payloads, and robust connectivity possibly enhanced by satellite relays. These technical constraints guide our analysis of LPWAN technologies, which inherently support such features and are increasingly integrated into cellular chipsets and IoT ecosystems.

The remainder of this paper is organized as follows. Section II presents a comparative analysis of Low Power Wide Area Network (LPWAN) technologies and evaluates their suitability for disaster communication scenarios. Section III discusses the integration of satellite-based communication, with a focus on emerging Direct-to-Satellite (DtS) IoT architectures. Section IV outlines the proposed smartphone-based Disaster Mode framework, detailing its connectivity logic and security mechanisms. Section V examines implementation challenges, energy efficiency considerations, and protocol-level interoperability. Section VI highlights policy recommendations and the potential for large-scale deployment. Finally, Section VII concludes the paper with a summary of findings and future research directions.

II. STUDY OF EXISTING OR ONGOING EFFORTS:

Many recent initiatives explore resilient emergency networks using LPWAN technologies. For example, Dryad Net works has deployed Silvanet, a mesh of LoRaWAN sensors in wildfire-prone forests. These solar-powered sensors detect fires and forward alerts over a multihop LoRaWAN mesh; a satellite uplink (via Swarm) carries data from remote gateways back to a control center [20]. This system operates even where cellular networks are down, illustrating the potential of LPWAN+satellite integration. Researchers have also proposed LoRa-based disaster networks: Schmidt et al.'s “BPoL” design uses LoRa with delay tolerant networking (DTN) to bridge connectivity gaps in catastrophes [25]. Likewise, Pham et al. showed via simulation that a multihop LoRa mesh can effectively expand coverage for emergency messaging [26]. In the crisis informatics community, LoRaWAN's attributes (long range, low power, low cost) have attracted attention [27]. Kuntke et al. note that LoRaWAN's “low energy consumption and high range” make it well-suited for post-disaster communications, such as restoring links after infrastructure breaks or supporting environmental warning sensors. Other LPWAN efforts include a Sigfox-based fire detection network using smoke and gas sensors, and an LPWAN+SD WAN hybrid system proposed by Cheimaras et al. to provide off-grid emergency services [24].

While prior efforts have demonstrated the feasibility of LPWAN-based systems for emergency communication often using dedicated external modules, fixed sensors, or custom deployments, there remains a significant gap in smartphone-based solutions. Existing studies typically do not address how LPWAN technologies can be integrated directly into mobile devices or used with satellite fallback in real-world disaster scenarios. This paper uniquely evaluates the feasibility of enabling a native "Disaster Mode" in smartphones, which autonomously switches to LPWAN or satellite connectivity when terrestrial networks are unavailable. It also provides a comparative analysis of LPWAN standards and discusses relevant policy and infrastructure considerations necessary for scalable deployment.

III. LOW POWER WIDE AREA NETWORKS (LPWAN):

Low Power Wide Area Networks (LPWAN) represent a class of wireless communication technologies designed to provide long-range connectivity while consuming minimal power. These networks are specifically engineered for applications that require the transmission of small amounts of data at infrequent intervals. LPWAN protocols enable devices to operate over several kilometers, even in challenging environments, while maintaining extremely low energy consumption. This makes LPWAN ideal for battery-operated devices that must function for years without maintenance [13]. The primary value of LPWAN lies in connecting distributed devices efficiently where traditional networks are impractical or expensive. It's not built for high-speed data, but excels in use cases like sensor telemetry, infrastructure monitoring, and especially emergency alerts under constrained conditions. LPWAN technologies are already used in sectors such as agriculture (remote sensing, irrigation), smart cities (utility metering, traffic monitoring), and industrial IoT. But one of their most impactful uses is in disaster response, where they can deliver emergency alerts and sensor updates when other systems fail [14]. Common LPWAN standards include LoRa, Sigfox, and Narrowband IoT (NB-IoT). Though they differ in architecture and spectrum, they all aim to enable energy-efficient, long range, low-cost wireless communication.

IV. TECHNICAL COMPARISON OF LORA, SIGFOX, AND NB-IOT:

This section compares three key LPWAN standards, LoRaWAN, Sigfox, and NB-IoT, for use in Disaster Mode, as extensively evaluated in recent industry comparisons [1][8]. Each is evaluated based on factors like spectrum usage, coverage, power efficiency, data rate, infrastructure needs, and potential for satellite integration. All these technologies rely on a gateway or satellite link to transmit messages from devices to emergency systems, using either terrestrial networks or satellite relays based on availability.

A. LoRaWAN:

LoRa (Long Range) is a proprietary modulation technology by Semtech that operates in unlicensed sub-GHz ISM bands (e.g., 868 MHz in Europe, 915 MHz in North America). LoRaWAN is the network protocol built on top of the LoRa physical layer. LoRaWAN enables long-range communication up to 15–20 km in rural areas and 2–5 km in urban environments, by adjusting its spreading factor to trade data rate for sensitivity [13]. Its link budget can exceed 150 dB, making it resilient in obstructed environments. The hardware is energy-efficient: LoRa chips typically consume under 100mA during transmission, and less than 1A in sleep mode [4]. Class A devices, which are most power-efficient, transmit only when needed and listen for downlink responses immediately after.

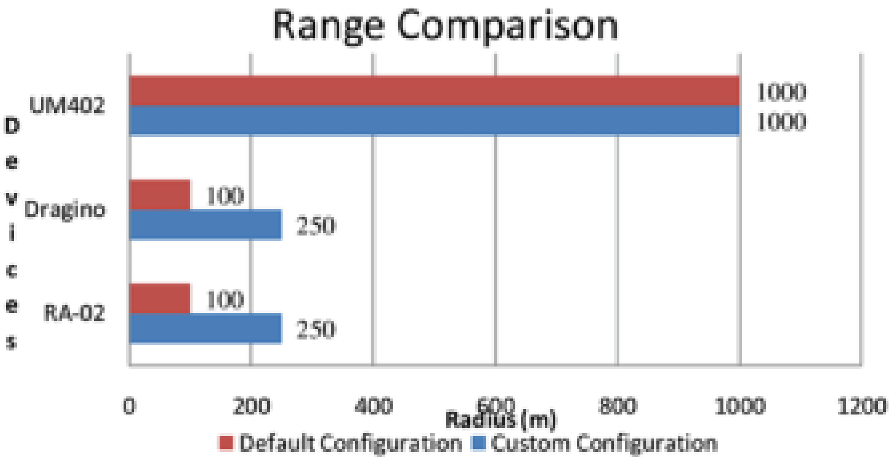

Fig. 1. Range Comparison graph between Lora devices.

As shown in Fig. 1, custom configurations significantly improve the communication range of RA-02 and Dragino modules, while the UM402 device consistently achieves a 1000-meter range in both setups.

B. Sigfox:

Sigfox is a centralized LPWAN solution operating in unlicensed ISM bands using ultra-narrowband modulation. It limits transmissions to very small payloads (12 bytes max), with a duty cycle of 140 uplinks and 4 downlinks per day. Sigfox achieves long range up to 40 km in rural areas, due to high receiver sensitivity (–125 dBm). Transmit current draw is low (about 49mA at +14dBm), and sleep currents can drop to under 1A, making Sigfox devices extremely battery efficient [5]. However, its limitations include constrained data capacity, lack of uplink flexibility, and complete dependency on Sigfox managed infrastructure. Satellite support is planned via a 25 satellite constellation by Eutelsat [11].

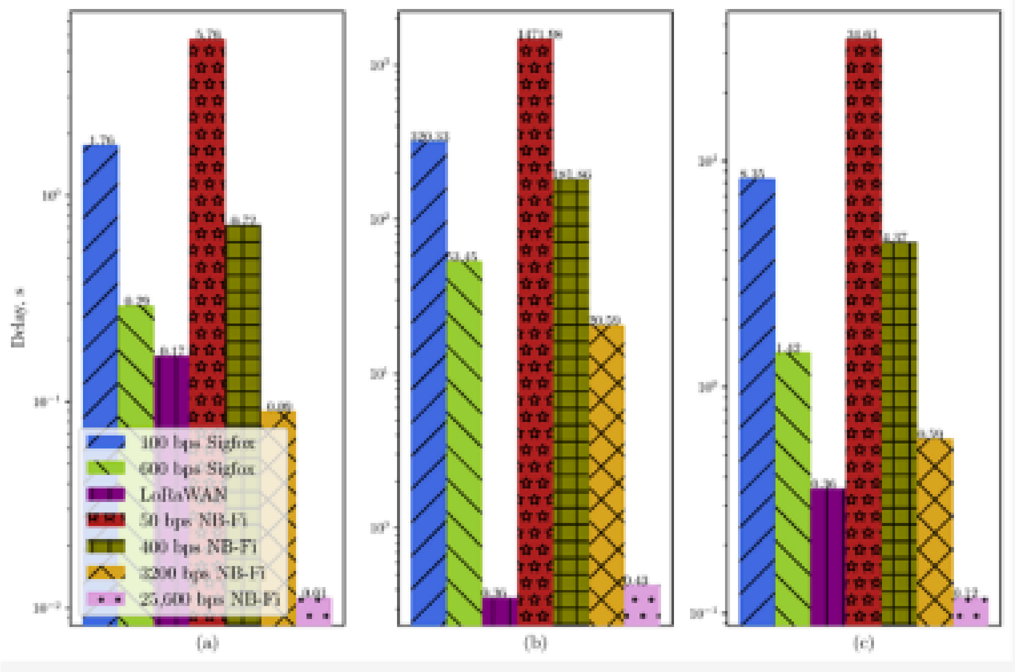

*Fig. 2. The average delay in (**a**) the unreliable transmission of short packets scenario; (**b**) the unreliable transmission of long packets scenario (**c**) the unreliable transmission of long packets scenario*

As shown in Fig. 2, Lower bit-rate technologies like B-Fi at 50 bps and Sigfox at 100 bps suffer from significantly higher delays, especially in long-packet reliable scenarios, while high-rate NB-Fi configurations (e.g. 25,600 bps) maintain sub-second latency.

C. NB-IoT

NB-IoT is a 3GPP standardized LPWAN protocol derived from LTE (introduced in Release 13). It operates in licensed spectrum [7] either in-band with LTE, in guard bands, or standalone. NB-IoT supports higher throughput (up to 250 kbps uplink), large payloads (1600 bytes), and very high link budgets (164 dB), enabling deep indoor coverage and strong signal resilience [6]. It consumes more power per transmission (up to 230mA), but power-saving features like eDRX and PSM modes enable multi-year battery life for low-frequency use,especially
when combined with emerging wireless power technologies [12][18]. Crucially, NB-IoT is increasingly built into modern smartphone base bands [21], making it highly suitable for Disaster

Mode. In India, BSNL's partnership with Skylo provides nationwide NB-IoT coverage via satellite backhaul, ensuring availability even in network deserts [10]

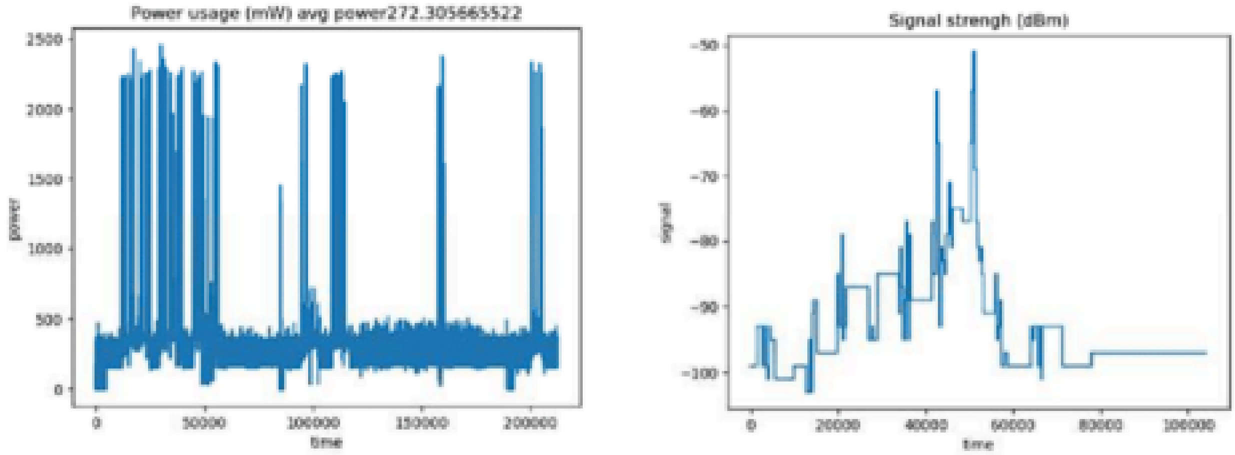

Fig. 3. NB-IoT power consumption vs signal quality (1 unit of time = 0.05 s).

As shown in Fig. 3, the power consumption and signal strength of NB-IoT over time. Spikes in power usage align with poor signal conditions, indicating energy is used to maintain connectivity. This highlights NB-IoT's trade-off between power efficiency and signal reliability.

V. DIRECT-TO-DEVICE SATELLITE CONNECTIVITY

Direct-to-Device (D2D) satellite communication allows smartphones to establish a direct RF link with Low Earth Orbit (LEO) satellites, eliminating the need for terrestrial base stations or gateway infrastructure. This capability forms the final fallback layer in the Disaster Mode hierarchy, ensuring connectivity when all conventional and LPWAN paths are unavailable.

In 2024, BSNL and Viasat launched India's first D2D satellite service, enabling low-data functions like SOS messages and UPI transactions via satellite. This aligns with 3GPP Release 17+ NTN specifications, which define support for satellite waveforms (S-band, L-band) within standard cellular chipsets.

Technically, D2D communication requires:

- Compact beamforming-capable antennas on the smartphone
- Support for Non-Terrestrial RATs (NTN-RAT) within the modem
- Line-of-sight access to the satellite (outdoor or near-or near-window usage)

This satellite path is activated only when the intelligent RAT logic determines that all terrestrial and LPWAN links have failed. Although D2D links have higher latency and power draw, they enable critical low-bandwidth communication in total infrastructure collapse scenarios.

Key challenges include RF power constraints on handheld devices, dynamic Doppler correction due to satellite motion, and regulatory approvals for shared spectrum use. Despite these, D2D satellite connectivity provides a resilient last-resort channel for emergency communication in

Disaster Mode.

While direct-to-satellite communication is ideal, not all LPWAN end devices, especially in obstructed or low-power environments can reliably reach LEO satellites. To address this, LPWAN repeaters or gateways equipped with satellite uplinks must be strategically deployed in vulnerable or disaster-prone regions. These intermediate nodes collect emergency messages from nearby LPWAN devices (e.g., LoRa, NB-IoT) and relay them to satellites, ensuring connectivity when cellular infrastructure fails. This indirect approach balances the energy efficiency of LPWAN with the reach of satellite networks and is especially critical where smartphones or IoT devices lack line-of-sight access to satellites.

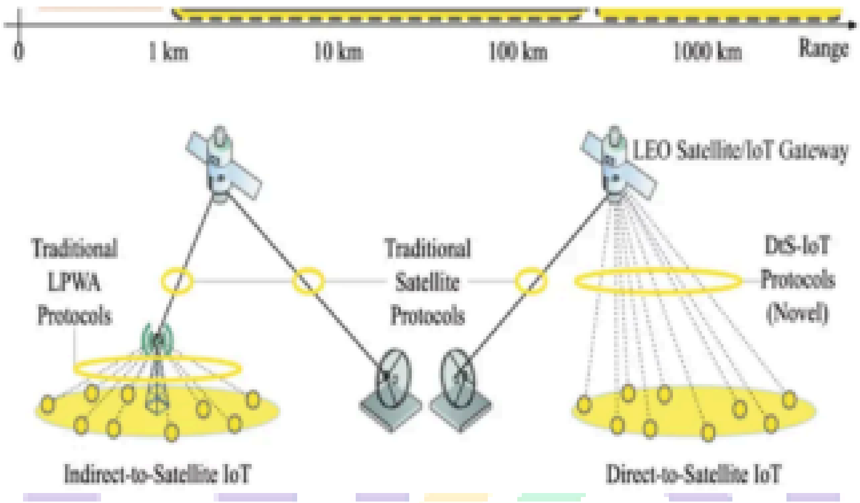
0
1 km
10 km
100 km
1000 km
Range
LEO Satellite/IoT Gateway
Traditional LPWA Protocols
Traditional Satellite Protocols
DtS-IoT Protocols (Novel)
Indirect-to-Satellite IoT
Direct-to-Satellite IoT

Fig. 4. *Comparison between indirect-to-satellite and direct-to-satellite (DtS) IoT architectures*

As shown in Fig. 4, there are two primary architectures for satellite-based IoT communication: indirect-to-satellite and direct-to-satellite (DtS). In the indirect model, IoT devices transmit data to nearby terrestrial gateways using LPWAN protocols like LoRa or NB-IoT. These gateways then forward the data to satellites via traditional satellite protocols. While widely adopted, this method depends heavily on ground infrastructure, which may fail during disasters. In contrast, the DtS model enables devices to communicate directly with LEO satellites using novel DtS-IoT protocols, eliminating the need for intermediary infrastructure. This approach is particularly valuable for emergency communication systems where network independence and wide-area coverage are critical.

VI. Smartphone-Based Disaster Mode: Architecture, Connectivity Logic, and Security Measures

The Disaster Mode system uses a layered communication architecture that enables smartphones to switch between multiple connectivity options like LTE/5G, NB-IoT, LoRa, and satellite, based on availability. A software layer manages these radio technologies, ensuring that when an SOS is triggered, the device attempts to send the message through the most efficient available path, with satellite communication as the final fallback. Messages include essential data like location and distress type and are relayed to emergency platforms for response.

As shown in Fig. 5, the Disaster Mode system follows a hierarchical decision flow to determine the best available communication channel during emergencies. When triggered, the system first checks for active LTE or 5G connectivity and, if unavailable, sequentially evaluates LPWAN (e.g., NB-IoT or LoRa) and satellite options. If no viable network is found, the emergency message is temporarily logged locally and retransmission is attempted later. This intelligent fallback mechanism ensures that distress signals are sent with minimal latency and maximum power efficiency, even in network-compromised disaster zones.

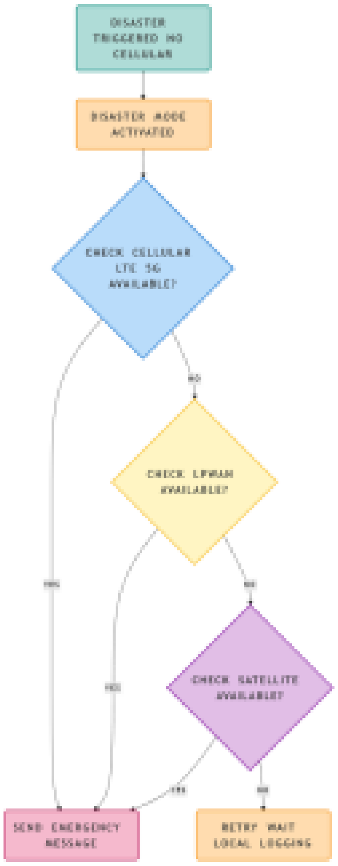
DISASTER TRIGGERED NO CELLULAR
DISASTER MODE ACTIVATED
CHECK CELLULAR LTE 5G AVAILABLE?
NO
CHECK LPWAN AVAILABLE?
YES
NO
YES
CHECK SATELLITE AVAILABLE?
YES
NO
SEND EMERGENCY MESSAGE
RETRY WAIT LOCAL LOGGING

Fig. 5. Decision Flow for Smartphone-Based Disaster Mode Communication

An **intelligent Radio Access Technology (RAT) selection system** continuously evaluates network conditions, such as signal strength, latency, and energy cost and dynamically switches to the optimal communication mode. The system prioritizes LTE/5G for speed, NB-IoT for wide coverage, LoRa for low energy use, and satellite only when no other option is viable. This decision-making logic is based on 3GPP Release 17 standards and runs autonomously within the device.

To ensure security and prevent misuse, the system includes safeguards against spoofing and denial-of-service (DoS) attacks. These include digital signatures, encryption, rate limiting, user confirmation prompts, and centralized validation. Together, these measures ensure that only authenticated distress messages are delivered, preserving system trust and integrity during emergencies.

VII. Conclusion

We have outlined a comprehensive Disaster Mode concept for smartphones using LPWAN connectivity when cellular and Wi-Fi fail. Our analysis incorporated authoritative data and test results to compare LoRaWAN, Sigfox, and NB-IoT. LPWAN links indeed provide ultra-low-power, long-range connectivity for emergency messaging. Our detailed comparisons (summarized in tables and figures) reveal that NB-IoT stands out as the preferred technology under most conditions: it balances wide coverage, modest power use, and higher throughput. Crucially, with BSNL's forthcoming satellite-enabled NB-IoT networks and direct to device satellite services, disaster communication can be extended nation- or globe-wide. Thus, we recommend that smartphone manufacturers and standards bodies incorporate NB-IoT for emergency Disaster Mode, to ensure that when catastrophe strikes, even the last mile of rescue messages can get through.

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